

Name _____

Date Feb 16, 2024

Total
16.5
/17

1. Use the graph of the function $f(x)$ to evaluate each expression.

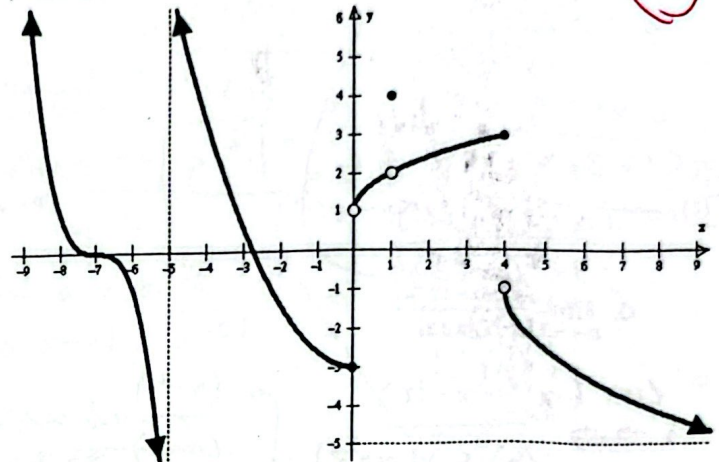
(2)

a) $f(1) = 4$

b) $\lim_{x \rightarrow 0^+} f(x) = 1$

c) $\lim_{x \rightarrow 4^-} f(x) = 3$

d) $\lim_{x \rightarrow -5} f(x) = \text{DNE}$



2. Rationalize the denominator of the expression. Simplify fully.

(2)

a) $\frac{4\sqrt{2}+5}{6-4\sqrt{2}} \times \frac{6+4\sqrt{2}}{6+4\sqrt{2}}$

$$= \frac{31 + 22\sqrt{2}}{2}$$

$$= \frac{(4\sqrt{2}+5)(6+4\sqrt{2})}{6^2 - (4\sqrt{2})^2}$$

$$= \frac{24\sqrt{2} + 32 + 30 + 20\sqrt{2}}{36 - 32}$$

$$= \frac{62 + 44\sqrt{2}}{4}$$

3. If $\lim_{x \rightarrow 2} f(x) = 4$ and $\lim_{x \rightarrow 2} g(x) = 7$, use all applicable properties of limits to evaluate the following limit.

1.5

(2)

$$\lim_{x \rightarrow 2} \left[\frac{\sqrt{7g(x) - 6x^2}}{(f(x))^2 - 1} \right]$$

$$= \lim_{x \rightarrow 2} \left(\frac{\sqrt{7g(x) - 6x^2}}{(f(x))^2 - 1} \right)$$

$$= \left(\frac{7 \lim_{x \rightarrow 2} g(x) - 6 \left[\lim_{x \rightarrow 2} (x) \right]^2}{\left(\lim_{x \rightarrow 2} f(x) \right)^2 - 1} \right)^{1/2}$$

$$\left(\frac{7(7) - 6(2)^2}{(4)^2 - 1} \right)^{1/2}$$

$$= \frac{25}{15}^{1/2} = \frac{5}{3}$$

$$= \frac{25}{15}^{1/2}$$

$$= \frac{5}{3}$$

$$= \frac{1}{3}$$

5.5

4. Evaluate the following limits. State if the limit does not exist.

$$\begin{aligned} \text{a. } \lim_{x \rightarrow 3} \frac{\frac{1}{3} - \frac{3}{x^2}}{x-3} &= \lim_{x \rightarrow 3} \frac{\frac{x^2 - 9}{3x^2}}{x-3} \\ &= \lim_{x \rightarrow 3} \frac{x^2 - 9}{3x^2} \cdot \frac{1}{x-3} \\ &= \frac{(x+3)(x-3)}{3x^2(x-3)} \\ &= \frac{x+3}{3x^2} \\ &= \frac{(3)+3}{3(3)^2} \\ &= \frac{6}{27} = \frac{2}{9} \end{aligned}$$

$$\begin{aligned} \text{b. } \lim_{x \rightarrow \frac{5}{2}} \frac{4x^2 - 20x + 25}{2x^2 + 7x - 30} &= \lim_{x \rightarrow \frac{5}{2}} \frac{(2x-5)(2x-5)}{(2x-5)(x+6)} \\ &= \frac{(2x-5)}{(x+6)} \\ &= \frac{2(\frac{5}{2}) - 5}{(\frac{5}{2} + 6)} \\ &= \frac{5 - 5}{\frac{17}{2}} = 0 \end{aligned}$$

$$\begin{aligned} \text{c. } \lim_{x \rightarrow -3} \frac{x^2 - x - 12}{(x-5)|x+3|} &= \lim_{x \rightarrow -3} \frac{(x-4)(x+3)}{(x-5)|x+3|} \\ &= \frac{(x-4)}{(x-5)} \quad \text{if } x > -3 \quad \text{RS} = \frac{2(\frac{5}{2}) - 5}{(\frac{5}{2} + 6)} \\ &= \frac{(x-4)}{(x-5)} \quad \text{if } x < -3 \quad \text{LS} = \frac{-(x-4)}{(x-5)} = \frac{-(-3-4)}{(-3-5)} = \frac{7}{-8} = -\frac{7}{8} \\ &\therefore \text{DNE} \end{aligned}$$

6. The tadpole population (P), in thousands, in Lake Ontario at a time, in days is modelled by the function

$$P(t) = \begin{cases} 2t^2 + 1 & 0 \leq t < 2 \\ 9 - \frac{3}{2}t & 2 \leq t < 6 \end{cases}$$

This function describes a sudden change in the population at $t = 2$, due to a storm.

a. Sketch the graph of $P(t)$.

b. Evaluate $\lim_{t \rightarrow 2} p(t)$. Explain.

$$\begin{aligned} \lim_{t \rightarrow 2} p(t) &= \text{DNE} \quad \text{the LS and RS limits do not equal each other @ } x=2 \\ \lim_{t \rightarrow 2^-} p(t) &\neq \lim_{t \rightarrow 2^+} p(t) \quad \therefore \text{the limit DNE} \end{aligned}$$

c. Determine approximately how many blue jays were killed by the storm.

$$\begin{aligned} &= 2(2)^2 + 1 = 9 \\ &= 9 - \frac{3}{2}(2) = 6 \end{aligned}$$

$$= 9 - 6$$

$$= 3 \rightarrow 3000$$

